MRITYUNJAY GUPTA

Bhilwara, India | mrityunjay.coder@gmail.com | +91 95879 13732 | mrityunjaygupta.com | https://github.com/mj-112358

EDUCATION – Jayshree Periwal International School, Jaipur

- **IBDP predicted: 45/45 (Grade 11: 42/42); Higher Level:** Computer Science, Mathematics: Analysis and Approaches, Physics, Chemistry. **Standard Level:** English A: Language and Literature, French B.
- **SAT: 1550** (790 Mathematics; 760 Evidence-Based Reading and Writing)
- **IGCSE: 10/10 A***; **Cambridge Outstanding Learner Award, Top in the World** in Mathematics (Extended); school highest in Computer Science, Chemistry and French; Top 1% in Business Studies.

ACADEMIC HONORS & AWARDS

- **Regeneron ISEF 2026, Third Grand Award:** Recognized for EcoFog, passive fog-harvesting mesh; One of the 3 Indian Grand Award Winners in 2026; One of 2 Indian Third-Grand-Awardees in 2026; First ever from my school
- **2x S.T. Yau High School Science Award (Asia) Finalist:** Math & Physics (Top 12 & Top 10); Chemistry Semifinalist
- **The Diana Award:** Recognized for social entrepreneurship through AmbuGo and PredictX
- **International Children's Peace Prize, Top 30:** Recognized for democratizing access to education in tribal communities
- **$10,000 Leaders of Today Award** (Entrepreneurship & Innovation Track)**:** Only Recipient from India
- **Youth Parliamentarian** at the **UNGA Science Summit 2026**: Contributed to drafting policies for finance & regulation of scientific innovation
- **University of Rochester** Bausch + Lomb Honorary Science Award (~$20K scholarship)
- **Forbes India**, ***Visionaries Empowering India's Next Decade***: Youngest Entrepreneur recognized

TECHNICAL & ENTREPRENEURIAL PROJECTS | AI & HEALTH ENGINEERING

Co-Developer, AmbuGo | Ambulance Prioritization | Github | Paper & Simulations *Grades 9–12*

- Engineered a retrofit-compatible ambulance priority system using Bayesian arrival-time estimation and risk-calibrated route selection; recorded **32% reduction in emergency response time in Phase 0 (Pilot Testing)**.
- Formulated a **minimum-intervention traffic-control problem** that jointly selects ambulance routes and signal overrides while enforcing finite road storage, queue propagation, legal phase transitions and post-passage recovery.
- Evaluated the controller through 14,400 mesoscopic and 1,728 microscopic traffic simulations; achieved a 44.6% reduction in simulated ambulance duration versus routing alone, retaining **94% of aggressive priority's time savings** while reducing its congestion exposure by 5.8%.
- Deployment with Punjab traffic authorities covering **121 intersections** and prioritization of **700+ ambulance movements per day**.

AI Software Lead, PredictX | Low-Cost Disease Screening | Github | Provisional Patent *Grades 10–12*

- Developed an integrated AI architecture assessing five diseases using routine blood and urine biomarkers. Engineered **compatibility-controlled preprocessing**, reducing redundant laboratory conversions by 80% while preserving independently trained disease models. Evaluated five models on held-out cohorts, including AUROCs of 0.923 (diabetes), 0.760 (liver), 0.785 (cardiovascular), and 0.922 (breast cancer).
- Extended this research by developing an **Airflow Obstruction Screening extension** that reconstructs spirometry flow-volume curves (for COPD) with physics-informed obstruction models, checks maneuver quality and studies when incomplete tests warrant repeat measurement. **Reduced missed airflow-obstruction cases by 36 percentage points (42.7% to 6.7%)**. (Click to View Paper)
- Tested for **3,000+** patients at 5 hospitals; **Standard tests corroborated suspected conditions in ~65% of high-risk cases**. Secured R&D grant of **~US$20,000 (INR 20,00,000)** for advancing the physics-informed respiratory-screening research

Developer & First Author, AERIS | Accessible Speech and Emotion Recognition | Paper *Grades 9–10*

- Developed an affordable assistive wearable that translates spoken emotion cues into distinct tactile signals. Integrated MFCC-based speaker classification, a 1D convolutional network, pretrained Wav2Vec2 emotion recognition and real-time haptic feedback. Published first-author research in IEEE ICESC 2025; **99.3% classification accuracy**; **250 ms latency**.

RESEARCH | ENGINEERING, COMPUTING & PHYSICAL SCIENCES

Researcher, EcoFog | Passive Fog Collection and Multiscale Materials Optimization | Paper *Grades 10–12*

Research guidance: Dr. Kaushik Pal, IIT Roorkee via IRIS

- Addressed a central fog-harvesting trade-off: dense fibers capture droplets but can clog the mesh. Designed a coplanar architecture pairing **electrospun PVA/PEO capture nanofibers** with a **3D-printed PLA drainage frame**; tuned **TiO_2/SiO_2 loading**, requiring no specialized infrastructure in deployment.

- Achieved **17.3 ± 2.1 L/m²/day** outdoors, **3.46× standard** Raschel mesh, with **~50% less retained water**; verified fiber structure and surface chemistry through FESEM, FTIR, XRD and contact-angle testing.

Co-Researcher, Musical Timbre Modeling | Complex Analysis and Signal Compression | Paper *Grades 11–12*

Research guidance: Mr. Jitendra Soni, HOD, Math & Mr. Vijay Jharia, Math Faculty @ JPIS

- Derived a **phase-aware mathematical model** that reconstructs an instrument's evolving harmonic spectrum from a small set of complex singularities, capturing how individual frequencies change in strength and phase over time. Established conditions under which these parameters can be uniquely recovered from spectral measurements.
- Developed a **variable-projection optimization method** that separates linear amplitude estimation from nonlinear singularity fitting, using matrix-pencil initialization to recover model parameters. Preserved **93.6% of harmonic energy using 170 instead of 1,088 real parameters** across 72 instrument tones; predicted unseen harmonics with **1.5% median relative error** in synthetic tests.

Co-Researcher, Vibration-Driven Rope Slip | Predictive Contact Mechanics | Paper *Grades 11–12*

Research guidance: Mr. Gajendra Sethi, HOD, Physics @ JPIS

- Derived a **closed-form logarithmic equation** predicting where a vibrating rope transitions from slipping to gripping around a cylinder. Combined capstan friction theory with calibrated contact stiffness to model how the boundary shifts with vibration amplitude and rope tension.
- Developed an **optical motion-tracking and regression pipeline** to measure rope slip across four wraps; compared five mathematical models using Bayesian Information Criterion to identify stable slip boundaries. Validated across **84 physical trials (0.117-rad RMSE)**, with measured tension sensitivity matching theoretical predictions.

PROFESSIONAL EXPERIENCE | AI, ROBOTICS & SOFTWARE

Business Growth & ML Intern, Human Archive | Robot Learning | Click to View My Work *Grade 11*

- Bridged factory-floor operations and robotics research by translating real-world manufacturing activities into structured human-action demonstrations for embodied-AI training.
- Trained and evaluated physical robots on everyday manipulation tasks, iterating demonstrations and data-quality checks in response to failed grasps and task executions to strengthen the transfer of recorded human actions to robot policies. Sold to 6 AI Labs.

Web Developer for Prof. Femida Handy (Penn SP2) | Academic Portfolio Website | femidahandy.com *Grade 12*

- Met Professor Handy at a Penn Forerunner after my acceptance to the M&T Summer Institute; our conversations about her research on philanthropy, the same field behind my own tribal-education fundraising, grew into building her professional website.
- Delivered the final website within two weeks, with every round of her feedback incorporated

CTO, Wink | On-Device Retail Intelligence and Privacy-First SaaS | Promo Video *Grade 9*

- Built a retail behavior engine using YOLO tracking, person re-identification and point-in-polygon zone analytics to measure where shoppers stop, how long they stay and how movement changes across a store.
- Shifted camera inference to retailers' own computers, reducing monthly costs from ~$500 to $5 and keeping raw footage in-store.

LEADERSHIP | COMMUNITY & EDUCATION

Deputy Head Boy, Student Council | Student Leadership | School of 2,550 *Grades 11–12*

- After winning the school's first ISEF Grand Award, became an informal IRIS research coach: helped 40 student projects shape research questions, troubleshoot experiments, critique evidence and turn early ideas into defensible projects; 17 advanced to IRIS Nat'l Fair (Shortlist for Team India, Regeneron ISEF); led five AI workshops.

President, Musica Universalis | Making Music Education and Credentials Accessible *Grades 9–12*

- Began by teaching guitar, then built accessible digital lessons as technology head and became president; now coordinating music learning with five NGO partners across six countries for 5000+ children.

Finance & Technology Head, Students for Last-Mile Learning | Tribal Education Access *Grade 11*

- Led Fundraising efforts to raise $185,000 for Ekal Vidyalaya (Education NGO), supporting one year of education for 16,500 tribal students across 550 Ekal schools. Set a world record for the largest student-led tribal-education fundraiser.

TECHNICAL SKILLS

Programming: Python, C++, SQL, Git, Linux
Frameworks: PyTorch, scikit-learn, Hugging Face, OpenCV, NumPy, pandas
AI Techniques: Machine Learning, Deep Learning, Computer Vision, Speech & Audio Processing, Physics-Informed ML, Edge AI
Foundations: Probability & Statistics, Optimization, Signal Processing, Algorithms & Data Structures
Deployment & Research: FastAPI, Streamlit, 3D Printing, Electrospinning